Senior Data Engineer

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PROFESSIONAL SUMMARY:

- Over 9+ years of extensive background as a Data Engineer with a strong foundation in Data Analysis and Data warehousing, Data Governance and Data Security and Distributed Computing. Highly proficient in designing, building, and optimizing data architectures and pipeline for large scale data processing.
- Well-versed in working with Big Data technologies, including Hadoop, Apache Spark, PySpark, Hive, HDFS, Kafka and various NoSQL databases.
- Skilled in utilizing AWS services such as S3, EMR, EC2, Glue, Lambda, IAM, Kinesis, RDS, Route 53, VPC, CodeBuild, CodePipeline, CloudWatch, and CloudFormation to architect scalable and secure data solutions.
- Strong capability in Python programming for data science, along with the Django framework for analytics applications and ETL (Extract, Transform, Load) processes.
- Demonstrated Knowledge in Provisioning and managing AWS resources such as EMR clusters, S3 buckets, EC2 instances, and RDS databases.
- Adept in data transfer from AWS S3 to Redshift using Informatica tools and daya integration processes.
- Strong programming knowledge in Java, Python, SQL, T-SQL, and R.
- Thorough understanding of distributed storage architectures and parallel computing for large-scale data processing.
- Strong familiarity with the full Data lifecycle, including Data Cleansing, Data Conversion, Data Mapping, Performance Optimization, and System Validation, Data Governance policies.
- Competent in writing complex SQL queries and developing database objects such as stored procedures, triggers, packages, and functions using SQL and PL/SQL.
- Enhanced performance of PySpark jobs on Kubernetes clusters for faster processing.
- Supported ad-hoc business needs by developing complex stored procedures using tools like TOAD.
- Thorough grounding in Python programming for data manipulation and automation.
- Solid grasp of Agile/Scrum and Waterfall methodologies for managing and executing projects.
- Collaborated closely with business users and senior management to implement technical solutions aligned with business goals.
- Strong foundation in Big Data processing, covering data ingestion, storage, querying, and analysis using Hadoopand related tools.
- Created MapReduce programs for processing large datasets, managing diverse file formats such as Text, JSON, XML, Parquet, and Avro.
- Extensive involvement in data modeling, data governance, and delivering technical solutions in enterprise environments.
- Practical knowledge of NoSQL databases like MongoDB and Cassandra.
- Competent in normalization and de-normalization techniques, including work on 3NF in both relational and dimensional data models.
- Deep understanding of Kimball (Dimensional) and Inmon (Relational) modeling methodologies.
- Capable of using MS Excel and MS Access for data analysis and meeting business reporting requirements.
- Strong skills in Data Analysis and translating business needs into technical designs.
- Applied PySpark for reading and writing data in various formats like JSON, ORC, and Parquet on HDFS.
- Practiced in using Sqoop for moving data between HDFS, Hive, and Relational Database Systems as per requirements.
- Expertise with Spark to boost performance through SparkContext, SparkSQL, and Spark Streaming.
- Strong foundation in data warehousing, particularly in creating Slowly Changing Dimensions (SCD) tables and providing data migration services.
- Skilled in using BI(Business Intelligence) tools like Tableau, Power BI, and Cognos for Data Visualization and creating insightful dashboards.

TECHNICAL SKILLS:

Programming Languages	SQL, Python, Java, Scala, Spark, R, UNIX.
Methodologies	RAD, JAD, System Development Life Cycle (SDLC), Agile.
Big Data Tools	Hadoop, MapReduce, Spark, Airflow, Nifi, HBase, Hive, Pig, Sqoop, Kafka, Oozie.
Databases	Oracle 12c/11g/10g, MY SQL, SQL Server, Teradata R15/R14, NO SQL (MongoDB, Cassandra, HBase).
Cloud Platform	AWS, Microsoft Azure, Google Cloud Platform (GCP)
AWS Cloud Management	Amazon Web Services (AWS)- EC2, S3, Redshift, EMR, Lambda, Athena, Glue, Kinesis.
BI Tools	SSIS, SSRS, SSAS Tableau, Power BI, Cognos.
ETL/Data warehouse Tools	Informatica 9.6/9.1, Tableau.
Distributed Computing	Hadoop, Apache Spark, Kafka.
API Integration	Developing and managing RESTful API services for data exchange.
Operating System	Windows, Unix, Sun Solaris.

PROFESSIONAL EXPERIENCE:

Client: Samsung ,Dallas, TX **May 2022 to Present**
Senior Data Engineer

Responsibilities:

- As a Data Engineer, supported leading the planning, construction, and execution phases within the Enterprise Analytics Team.
- Worked in the AWS environment for the creation and deployment of custom Hadoop applications.
- Designed and built ETL processes in AWS Glue to migrate campaign data from external sources like S3, ORC/Parquet/Text Files, into AWS Redshift.
- Led the architecture and design of data processing, warehousing, and analytics initiatives.
- Addressed and supported real-world business challenges using knowledge of Hadoop Distributed File Systems and open-source frameworks.
- Utilized AWS EMR to transfer large datasets (Big Data) into platforms like AWS data stores, Amazon S3, and Amazon DynamoDB.
- Created AWS Lambda functions using Python and Step Functions to orchestrate data pipelines.
- Took charge of enforcing data governance rules and standards to ensure consistency of business element names across different data layers.
- Participated in various stages of the development cycle, analyzing and shaping the system through Agile Scrum methodology.
- Conducted in-depth analysis of business problems and technical environments to inform the design and maintenance of data architecture solutions.
- Built a program using Python and Apache Beam, executing it in Cloud Dataflow for data validation between raw source files and BigQuery tables.
- Constructed data pipelines that enabled faster, more data-driven decision-making within the business.
- Performed data transformations in Hive and optimized performance using partitions and buckets.
- Tuned Hive queries to efficiently extract customer data from HDFS.
- Extensive background in using Elastic MapReduce (EMR) and configuring environments on Amazon AWS EC2instances.
- Handled scheduling with Oozie Workflow Engine to automate the running of multiple Hive jobs.
- Authored Spark scripts utilizing Python and Bash shell commands according to project needs.
- Wrote a Python program to manage raw file archival in GCS buckets.
- Implemented business logic by creating User-Defined Functions (UDFs) and configuring CRON jobs.

Environment: Hadoop 3.3, Spark 3.1, Python, Data Lake, GCS, HBase, Oozie, Hive, CI/CD, Big Query, Hive, Rest API, Agile Methodology, Code cloud, GCP.

Client: District of Columbia Access System, Washington, DC **Sep 2020 to April 2022**
Senior Data Engineer

Responsibilities:

- As a Data Engineer, led projects leveraging Spark, SQL, and AWS cloud infrastructure for efficient data handling and processing.
- Designed, deployed, and maintained comprehensive AWS infrastructure and services within a managed services environment.
- Worked on data governance strategies to establish operational frameworks in previously unregulated data environments.
- Participated in the ingestion, transformation, and computation of data using tools such as Kinesis, SQL, AWS Glue, and Spark.
- Migrated data from relational databases to NoSQL systems, providing an overarching view of the data distributed across different platforms.
- Designed and implemented end-to-end data solutions encompassing storage, integration, processing, and visualization using AWS services.
- Extensive involvement in migrating datasets and ETL workflows from on-premises systems to the AWS Cloudusing Python.
- Created PySpark pipelines using Spark DataFrame operations to load data into the Enterprise Data Lake (EDL), utilizing EMR for job execution and S3 as the storage layer.
- Designed and established an Enterprise Data Lake that enabled multiple use cases for analytics, data processing, storage, and reporting on large datasets.
- Collaborated with clients and solution architects to ensure high-quality data in source systems through activities like cleansing, transformation, and integrity maintenance in relational environments.
- Configured and set up self-hosted integration runtimes on virtual machines to facilitate access to private networks.
- Created visualizations and dashboards using the Power BI reporting tool to present actionable insights.
- Implemented Apache Airflow with AWS to orchestrate multi-stage machine learning pipelines involving Amazon SageMaker tasks.
- Built real-time streaming pipelines using Apache Spark and Python for efficient data flow management.
- Designed and implemented a Security Framework to manage fine-grained access control for S3 objects using AWS Lambda.
- Constructed PL/SQL packages, designed Database Triggers, and wrote user procedures, along with creating detailed user manuals for the new systems.
- Wrote User Defined Functions (UDFs) in Scala and PySpark to meet specific business logic requirements.
- Worked extensively with Informatica tools such as PowerCenter, MDM, Repository Manager, and Workflow Monitor.
- Assembled Kibana Dashboards and integrated multiple source and target systems with Elasticsearch for real-time analysis and end-to-end transaction tracking.
- Utilized REST APIs with Python to extract and load data from external sources into Google BigQuery.
- Employed Apache Spark for parallel data processing to significantly improve performance.
- Applied Python for web scraping and data extraction from various sources.
- Conducted multiple training sessions and demonstrations on Big Data technologies, sharing knowledge across teams.

Environment: AWS EMR, S3, RDS, Redshift, Lambda, Boto3, DynamoDB, Amazon SageMaker, Apache Spark, HBase, Apache Kafka, HIVE, SQ00P, Map Reduce, Apache Pig, Python, Tableau, UNICA, Kibana, Informatica.

Client: Royal Carrebian, Miami, FL **March 2018 to Aug 2020**
Data Engineer

Responsibilities:

- Worked as a Data Engineer, collaborating with Product Engineering team members to create, test, and support data-related initiatives.
- Gained a deep understanding of key business, product, and user questions to align data strategies with business needs.
- Followed the Agile methodology throughout the entire project lifecycle.
- Defined business objectives through discussions with stakeholders, functional analysts, and by participating in requirement collection sessions.
- Summarized the project's goals and highlighted business users' expectations from Business Intelligence (BI) and how they aligned with the overall project objectives.
- Led the estimation process, reviewed estimates, identified complexities, and communicated the outcomes to all stakeholders.
- Responsible for maintaining data consistency by enforcing data governance rules and standards across different layers.
- Migrated the on-premise environment to the cloud using Microsoft Azure.
- Involved in the migration of data from On-prem SQL Server to Cloud Databases like Azure Synapse Analytics (DW) and Azure SQL DB.
- Conducted data flow transformations using the Data Flow activity within Azure.
- Continuously monitored, automated, and enhanced data engineering solutions for optimized performance.
- Built pipelines, data flows, and carried out complex data transformations and manipulations using Azure Data Factory (ADF) and PySpark with Databricks.
- Drafted mapping documents to match source data columns with target data structures.
- Created Azure Data Factory (ADF) pipelines using Azure PolyBase and Azure Blob Storage.
- Performed ETL operations using Azure Databricks.
- Automated the ETL process by writing UNIX shell scripts.
- Worked on Python scripting to automate the generation of scripts, and conducted data curation using Azure Databricks.
- Utilized stored procedures, lookup functions, executed pipelines, data flows, copy data features, and Azure functions in ADF.
- Used Kaga to stream data from data sources and load it into HDFS for filtering.
- Created multiple Databricks Spark jobs using PySpark to perform table-to-table operations.
- Built visualizations and dashboards using the Power BI reporting tool for real-time insights.

Environment: Hadoop, Spark, Kafka, Azure Data Bricks, ADF, Python, PySpark, HDFS, ETL, Agile & Scrum meetings

Shakti Engineering Equipments | Hyd, India **July 2014 to Nov 2017**
Data Engineer

Responsibilities:

- Involved in the analysis, design, and implementation/translation of business user requirements.
- Worked on the collection of large sets of structured and unstructured data using Python scripts.
- Engaged in creating deep learning algorithms using LSTM and RNN architectures.
- Actively participated in designing and implementing data ingestion, aggregation, and integration processes within the Hadoop environment.

- Created Sqoop scripts for importing and exporting data from relational sources and managed incremental loading of customer and transaction data by date.
- Experience in setting up Hive tables, as well as implementing partitioning and bucketing strategies.
- Conducted data analysis and profiling using complex SQL queries across various source systems, including Oracle 10g/11g and SQL Server 2012.
- Identified inconsistencies in data collected from diverse sources to ensure data integrity.
- Designed the object model, data model, tables, constraints, and necessary stored procedures, functions, triggers, and packages for Oracle Database.
- Authored Spark applications for data validation, cleansing, transformations, and custom aggregations.
- Created custom aggregate functions using Spark SQL and executed interactive queries.
- Worked on installing clusters, commissioning and decommissioning data nodes, ensuring name node high availability, conducting capacity planning, and configuring slots.
- Built Spark applications for comprehensive batch processing using Scala.
- Stored transformed time-series data from the Spark engine, utilizing a Hive platform, into Amazon S3 and Redshift.
- Facilitated the deployment of multi-clustered environments using AWS EC2 and EMR, in addition to deploying Docker containers for cross-functional deployments.
- Visualized results using Tableau dashboards, and employed Python Seaborn libraries for data interpretation during deployment.
- Created PDF reports using Golang and XML documents to distribute to all customers at the end of each month.
- Applied various data mining techniques such as linear regression, logistic regression, classification, and clustering.

Environment: R, SQL server, Oracle, HDFS, HBase, AWS, MapReduce, Hive, Impala, Pig, Sqoop, NoSQL, Tableau, RNN, LSTM, Unix/Linux.